It will be especially critical for governments around the world to enact policies that drive rapid technological and economic transformations, including the full decarbonization of the global energy system. That's why Google advocates for stronger clean energy and climate policies, and partners with many other organizations to accelerate clean energy and climate progress. We've played instrumental roles in setting up advocacy organizations, including the Clean Energy Buyers Association (CEBA) in the United States, the RE-Source Platform in Europe, and the Asia Clean Energy Coalition (ACEC).

To learn more about our public policy and advocacy, including policy engagement, trade associations, memberships, and partnerships, see the Governance and Engagement section.

Challenges to address

While we're excited about our net-zero emissions journey, we recognize that we face many challenges. For example:

Growing energy needs

- As we work towards our absolute 50% emissions reduction target, our business is continuing to evolve to meet the needs of a rapidly digitalizing world, and to harness the opportunities presented by AI technology.
- Our operations and value chain are global and involve a diverse range of industries, which means we must make progress within hard-to-abate sectors and carbon-intensive geographies, such as the Asia Pacific region.
- We have a large number of different suppliers, both direct and indirect, each with climate programs that differ widely in maturity. In some cases, this limits our ability to successfully influence them, and to collect the data needed to accurately estimate our Scope 3 emissions.

Availability of carbon-free energy

- There are often long lead times between our investments and resulting GHG reductions. For example, with new wind and solar projects, it can take years before projects that we contract are constructed and begin to generate clean electricity.
- System-level decarbonization is proceeding more slowly than needed in many regions and sectors, and while we aim to accelerate these changes through our efforts and engagement, they aren't within our control.

Use of carbon removals

- Because the carbon removals sector is in the earliest stage of development, costs are currently high, and there are limited volumes of removals available for purchase.[68]

As we move towards our net-zero goal, the scale of these challenges will only increase, along with the difficulty of additional progress. We believe that the urgency of climate change demands immediate action, and setting ambitious, near-term targets is a valuable way to focus our efforts and drive important progress across the energy systems and value chains in which we operate.



Norther Offshore wind farm in Belgium (92 MW for Google)

Our approach

Our core business involves using data to solve hard problems, and our unique strength is combining engineering skills with an innovative mindset and a global reach.

There are three main focus areas that make up our approach to our net-zero goal: reducing emissions across our operations and value chain, advancing carbon-free energy, and addressing our residual emissions with carbon removals (see Figure 10).

FIGURE 10 **Our net-zero approach**



Carbon reduction

Reduce emissions across our operations and value chain



Carbon-free energy

Advance carbon-free energy across our operations and value chain



Carbon removals

Address our residual emissions with carbon removals

Measuring our carbon footprint

We began calculating our annual carbon footprint in 2006. Every year since 2009, we've publicly reported the results to CDP, a global organization that asks companies to disclose information on their GHG emissions performance and management.

In 2022, our total GHG emissions were approximately 10.2 million metric tons of carbon dioxide equivalent (tCO_2e), which represents our Scope 1, Scope 2 (market-based), and Scope 3 emissions. In 2022, our total operational emissions were approximately 2.9 million tCO_2e, where operational emissions represent Scope 1 and Scope 2 (market-based) emissions, as well as Scope 3 emissions from business travel and employee commuting, including teleworking (see Figure 11).

Scope 1 emissions

Scope 1 emissions represent 1% of our carbon footprint. Our main sources of Scope 1 emissions include transportation (company vehicles and aircraft), fuel use from back-up generators, natural gas use, and refrigerant leakage in our data centers and offices.

Scope 2 emissions

Scope 2 (market-based) emissions represent 24% of our carbon footprint, mainly due to the electricity demands of our data centers. Scope 2 emissions are thus a key focus of our decarbonization efforts, given that we have more direct control over our data centers than many other parts of our value chain. For over a decade, we've procured clean energy to reduce the carbon emissions from our electricity use: our Scope 2 (location-based) emissions, which don't take into account our renewable energy procurement through power purchase agreements (PPAs), represent 51% of our carbon footprint, demonstrating the dramatic emissions reductions that we've achieved through PPAs.

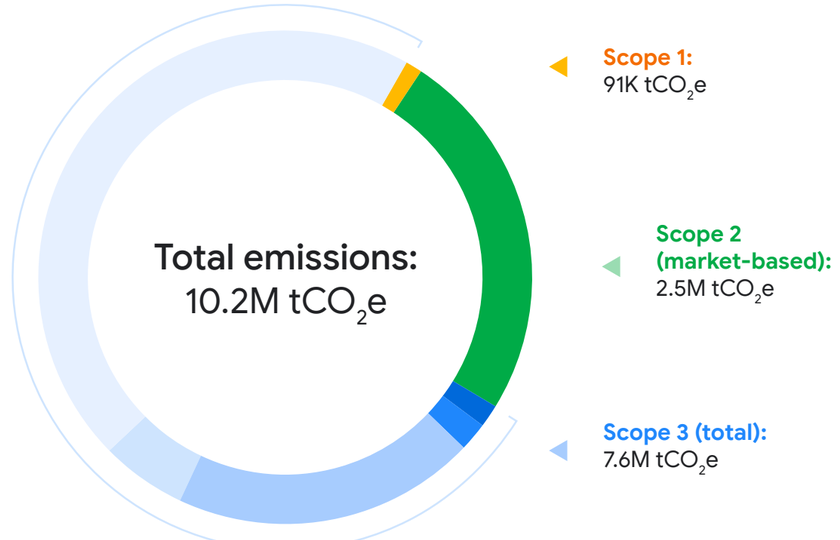
Scope 3 emissions

Our Scope 3 emissions are indirect emissions from other sources in our value chain, such as our suppliers, use of our consumer hardware products, and business travel. The majority of these emissions come from upstream manufacturing of consumer hardware products, the production of capital goods we have purchased for use in our data centers, and data center construction. After accounting for the impact of renewable energy procurement through PPAs, **Scope 3 emissions represent 75% of our carbon footprint.**

For more information on our GHG emissions, see the Reporting approach and methodology section and our Environmental data tables.

FIGURE 11

Our carbon footprint



Scope		tCO₂e	%
Scope 1		91,200	1%
Scope 2	(market-based)	2,492,200	24%
Scope 3	Business travel	211,000	2%
Scope 3	Employee commuting (including teleworking)	151,000	2%
Scope 3	Capital goods	2,096,000	21%
Scope 3	Upstream transportation and distribution	556,000	5%
Scope 3	Other categories	4,586,000	45%
Scope 3	(total)	7,600,000	75%
Total emissions		**10,183,400**	**100%**

Operational emissions: 2.95M tCO_2e

Reducing carbon emissions

We're working to reduce our emissions across our operations and value chain. Key efforts include energy-efficient and low-carbon facilities, electrification, sustainable travel and commuting, and supplier engagement.

Clean energy is also critical to reducing emissions across our value chain. Procuring carbon-free energy has a significant impact on reducing our Scope 2 emissions by eliminating emissions associated with our electricity use. In addition, a significant share of our Scope 3 emissions can be traced back to the electricity grids that power our suppliers and users, so accelerating the decarbonization of electricity grids is also key to our net-zero goal. For more information, see the Carbon-free energy section.

Our emissions reduction goal

In 2022, our total GHG emissions were approximately 10.2 million tCO_2e. We revised our Scope 3 methodology to calculate upstream emissions related to consumer hardware manufacturing by moving away from using a spend-based methodology to a Life Cycle Assessment (LCA)-based methodology.

We also made improvements to the quality of data used to estimate our emissions associated with manufacturing our data center hardware, as well as the LCAs and LCA emission factors used to calculate emissions related to data center construction.

These changes are reflected in our reported Scope 3 emissions for fiscal year 2022, but not for the prior years. Had we not made these changes, we estimate that our reported Scope 3 emissions would have been approximately 18% higher, remaining relatively flat when compared to the prior year (see Figure 12).

Our Scope 2 (market-based) emissions increased by 37%, primarily due to increased data center electricity consumption and a lack of full regional coverage of renewable energy procurement in the United States and Asia Pacific regions.

In 2022, we reported Scope 2 (location-based) emissions of 8.0 million tCO_2e, which doesn't take into account our renewable energy procurement, and Scope 2 (market-based) emissions of 2.5 million tCO_2e. This means that through our PPAs, in 2022, we achieved a 69% reduction in our emissions from our electricity use (see Figure 17).

Since 2011, our carbon intensity per unit of revenue has decreased by 76% while our business grew significantly over the same period.[69] We continue working to lower the correlation between our business growth and our GHG emissions.

TARGET

Reduce 50% of our combined Scope 1, 2 (market-based), and 3 absolute GHG emissions before 2030

2022 PROGRESS

Total GHG emissions:
10.2 million tCO_2e*

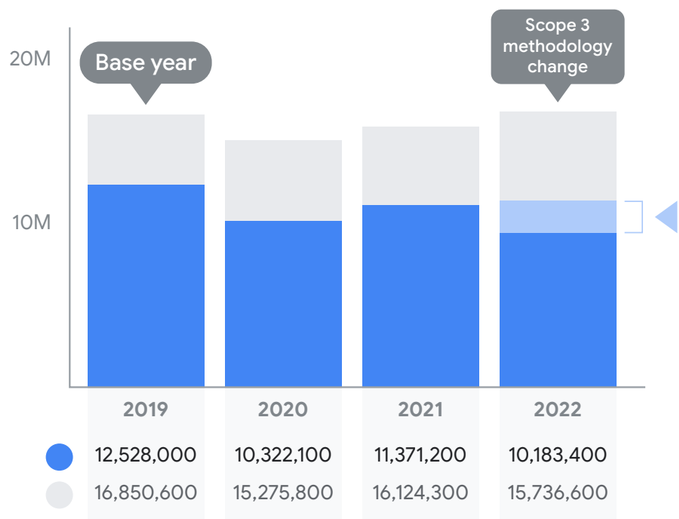
*Due to a methodology change, we're not reporting a percent progress towards this target for 2022

DETAILS

Year set: 2021 | Base year: 2019 | Target year: before 2030
Scope: Percent reduction in combined Scope 1, 2 (market-based), and 3 absolute emissions across our operations and value chain, when compared to a 2019 base year. This represents our operations and value chain, including consumer hardware products.

FIGURE 12

Annual emissions



(in metric tonnes of carbon dioxide equivalent [tCO_2e])

Legend

- Total emissions
- Business as usual (footprint in absence of PPAs)
- Estimated difference in 2022 total emissions due to Scope 3 methodology update

In 2022, our reported total GHG emissions decreased **due to improvements we made to our Scope 3 accounting methodology**, which we haven't yet applied to prior years.

Had we not made these methodology changes, we estimate that our reported 2022 Scope 3 emissions would have been approximately 18% higher, **remaining relatively flat when compared to 2021**.

Energy-efficient and low-carbon facilities

We pursue a set of integrated strategies to reduce the carbon footprint of our facilities, including energy efficiency, building and fleet electrification, on-site generation of clean energy, and carbon-free energy procurement. Reducing waste is another strategy, which is covered in the Circular economy section.

When designing our data centers and offices, we consider options to reduce their carbon impacts, such as the incorporation of low-GHG materials and adaptive reuse of existing buildings. For example, in Sunnyvale, California, we're building our first ground-up mass timber building, which is projected to have 96% fewer embodied carbon emissions than an equivalent steel and concrete structure, factoring in sequestration.[70]

Preventing refrigerants from leaking and finding low-global warming potential (GWP) alternatives is critical for reducing global emissions.[71] We're taking steps to more accurately measure refrigerant leak rates and working to develop new technology solutions that can prevent emissions from these leaks. For example, at our data centers, we're developing and deploying cooling solutions that include natural, **low-GWP refrigerants**.

Google is a founding member of the iMasons Climate Accord, an industry coalition working to reduce GHG emissions across the technical infrastructure systems that support the digital economy. As a part of this effort, we're collaborating with industry peers to highlight the importance of low-GHG construction materials, such as greener concrete, and to accelerate progress in the development and deployment of these materials.



A visualization of the roof of our landmark Platform G office development in King's Cross, London.

Data centers

Google's data centers are the engine of our company, powering products like Gmail, Google Cloud, Search, and YouTube for billions of people around the world. We've worked to make Google's data centers some of the most efficient in the world, improving their environmental performance even as demand for our products has risen. We've done this by designing, building, and operating each one to maximize efficient use of energy, water, and materials.

In 2022, we filed a patent for using machine learning technology to improve our ability to prevent emissions from refrigerant leaks.

Our long-standing data center efficiency efforts are important because our data centers represent the vast majority of our direct electricity use. To reduce their energy use, we strive to build the world's most energy-efficient computing network, outfitting each data center with high-performance servers that we've custom-designed to use as little energy as possible. For example, our tensor processing unit v4 has enabled us to train machine learning algorithms with industry-leading efficiency, resulting in significant emissions reductions compared to business as usual.[72] We also install **smart temperature and lighting controls**, redesign how power is distributed to reduce energy loss, and employ **advanced cooling techniques,** including energy-efficient evaporative cooling.

At some sites, we also **recapture heat** generated by our servers and use it to heat our data center office spaces and other on-site facilities. We continue to assess opportunities

2022 HIGHLIGHT

Compared with five years ago, we now deliver approximately three times as much computing power with the same amount of electrical power.[73]

for off-site heat recovery, where environmentally sound and technically feasible, considering the remote location of many of our data centers and the infrastructure required. Heat recovery is also a key part of our office development strategy, as well as one of the main levers to help us electrify—because reusing heat often requires less energy than creating new heat.

Our efforts have paid off. On average, a Google-owned and -operated data center is **more than 1.5 times as energy efficient** as a typical enterprise data center[74] and, compared with five years ago, we now deliver approximately three times as much computing power with the same amount of electrical power.[75] In 2022, the average annual power usage effectiveness (PUE)[76] for our global fleet of data centers was 1.10, compared with the industry average of 1.55[77]—meaning that Google data centers use about 5.5 times less overhead energy for every unit of IT equipment energy. Since 2012, our average annual fleet-wide PUE has stayed at or below 1.12 (see Figure 13).

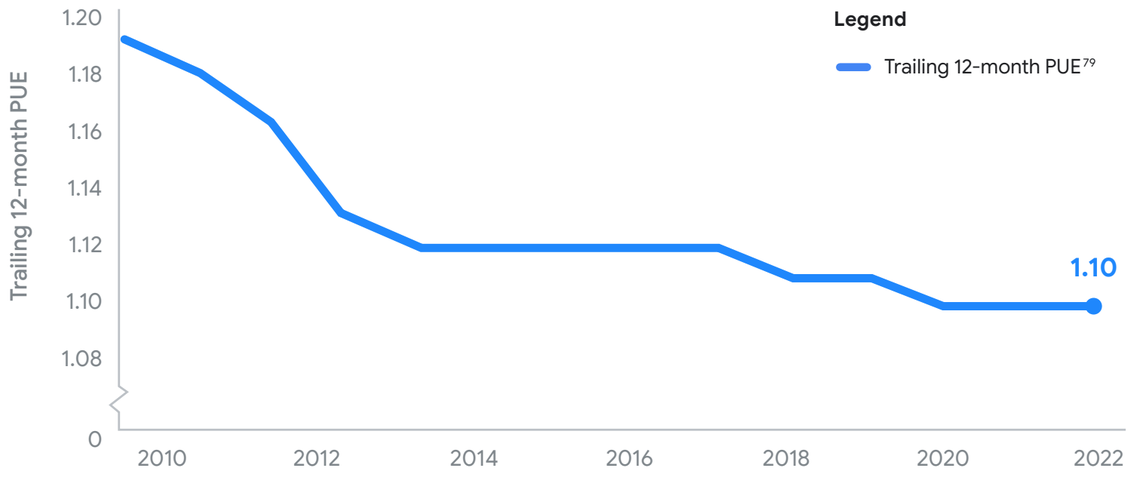
In 2022, we maintained our ISO 50001 certification for all Google-owned and -operated data centers in Europe that met our operational threshold for power usage. We were the first major internet company to achieve a multi-site energy management system certification to ISO 50001, which we first obtained in 2013.

2022 HIGHLIGHT

On average, a Google-owned and -operated data center is more than 1.5 times as energy efficient as a typical enterprise data center.[78]

FIGURE 13

Energy efficiency at Google-owned and -operated data centers



Historically, research has shown that as AI/ML compute demand has gone up, the energy needed to power this technology has increased at a much slower rate than many forecasts predicted. We have used tested practices to reduce the carbon footprint of workloads by large margins; together these principles have reduced the energy of training a model by up to 100x and emissions by up to 1,000x. For more information, see the AI for sustainability section.

Offices

For new office buildings and spaces, we optimize overall performance, including both efficiency and user experience (see Figure 14 on the next page for some examples). We design to internationally recognized and industry-leading **green building standards**, such as Leadership in Energy and Environmental Design (LEED) certification, Building Research Establishment Environmental Assessment Method (BREEAM), and International Living Future Institute's (ILFI) Living Building Challenge (LBC) and Zero Carbon Certification.

As of 2022, over 1.8 million square meters (nearly 20 million square feet) of Google office facilities have achieved LEED certification. Of our LEED-certified square footage, 27% has achieved a Platinum rating, and 60% a Gold rating.

As we work to improve performance across our existing portfolio, we use energy management strategies, building optimization, and data analytics to help drive efficiency measures like equipment upgrades, setpoint adjustments, and lighting retrofits.

2022 HIGHLIGHT

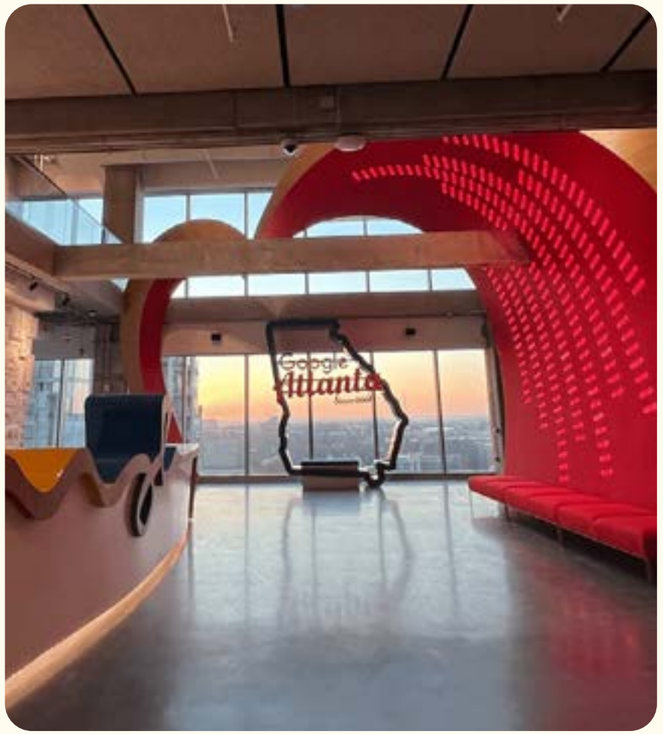
As of 2022, over 1.8 million square meters (nearly 20 million square feet) of Google office facilities have achieved LEED certification.



We received the highest LEED certification for our Google store in New York City's Chelsea neighborhood.

FIGURE 14

Examples of energy- and carbon-efficient Google offices



1105 Peachtree

in Atlanta, Georgia, U.S.A.

In 2022, we opened our 1105 Peachtree office, which is designed to meet LEED v4 Gold Certification. To promote low-carbon commutes, it's located near mass transit and offers on-site bike parking. It also includes smart building technology throughout to decrease energy use.



6 Pancras Square

in London, U.K.

Our 6 Pancras Square office was the first building project in the world to receive ILFI Zero Carbon Certification. It's also certified LEED v2009 Platinum and received a BREEAM Excellent rating.



Energy audits

in Asia-Pacific

In 2022, we conducted more than 20 energy audits across our largest Asia Pacific sites. As a result, we identified more than 200 energy conservation measures that we're now actively implementing—which are projected to save a total of 5,600 MWh of energy.

Electrification at our offices

Electrification—which helps enable transitioning away from fossil fuels—is an essential first step toward decarbonizing our operations, followed by 24/7 CFE procurement. We're exploring opportunities to **electrify our facilities**, from retrofitting existing building systems, to including electrification standards for new development, to working with landlords on electrification-friendly leases for buildings we don't own.

We've been piloting technologies to enable building electrification since 2010. For example, our first all-electric office—1842 North Shoreline in Mountain View, California—has been in operation for over a decade. Many of these pilots led us to confidently implement new technology in our large development projects. In 2022, we advanced a comprehensive plan to electrify systems across our building portfolio.

More recently, in 2022, we opened Bay View—an all-electric campus with an innovative geothermal installation and the largest electric kitchen in our global portfolio. Learn more in our Bay View spotlight.

We're also working to **eliminate use of natural gas** by designing all-electric kitchens at our offices, including in the Bay Area, Europe, and Latin America.

2022 HIGHLIGHT

In 2022, we opened Bay View—an all-electric campus.

Sustainable travel and commuting

Our transportation team plans, implements, and operates sustainable commuting options that help Googlers get to work by riding shuttles, carpooling, taking public transit, biking, or walking.

We have a growing number of **electric vehicles** in our corporate fleet, and the majority of our global fuel consumption from non-electric vehicles is renewable diesel. We strive to provide electric vehicle charging stations for 10% of the total parking spaces at our Bay Area headquarters, and we continue to work toward this design standard for new development projects.

To date, we've installed more than 5,000 electric vehicle charging ports at our offices. We estimate that employee EV commuting in the United States and Canada prevented roughly 12,000 tCO_2e emissions in 2022.[80]

Many campuses offer **commuter shuttles** to reduce individual vehicle trips. In 2022, we launched an expansion of our all-electric buses for the Bay Area and are introducing these throughout 2023 to replace our existing fleet. For 2022, our shuttle buses in the Bay Area produced savings of more than 10,000 tCO_2e emissions—the equivalent of avoiding more than 41 million vehicle km (25 million vehicle miles) or taking on average more than 2,000 cars off the road for a year.[81]

2022 HIGHLIGHT

Google shuttle buses in the Bay Area produced savings of more than 10,000 tCO_2e emissions—the equivalent of taking on average more than 2,000 cars off the road for a year.[82]

Supplier engagement

We work with many suppliers that are committed to sustainability, and we're partnering with them to develop decarbonization roadmaps and build essential data infrastructure to accurately quantify emissions and reductions across the value chain.

We engage with our suppliers—including hardware manufacturing and indirect services suppliers—to help reduce their energy consumption and GHG emissions, as stated in our Supplier Code of Conduct, which all suppliers are required to sign. We assess suppliers' practices to report, manage, and reduce their emissions and incorporate this into our supplier scorecard.

Reporting environmental data

We expect all our suppliers to report environmental data, and we encourage them to respond to CDP's Climate Change survey. In 2022, we invited 222 suppliers to participate, and at least 90% of our hardware suppliers, by spend, provided data. Our supplier data is used to help set goals and priorities for our sustainability program and we continue to refine our data quality and methodology for calculating the carbon footprint associated with the manufacturing of our hardware products throughout their life cycle.

In 2022, in addition to our CDP supply chain disclosure requests, we engaged with suppliers directly to drive improved data and accounting, including increased completeness and accuracy for their Scope 1, 2, and 3 emissions.

Developing roadmaps to reduce emissions

Since setting our net-zero emissions goal, we've engaged our key suppliers to develop roadmaps to reduce emissions across our supply chain. During Google's 2022 Supplier Sustainability Summit, we communicated the expectation for suppliers to publicly set targets to reduce GHG emissions. We provide in-depth training for key suppliers on setting ambitious GHG reduction and renewable electricity targets. For suppliers that completed our CDP supply chain survey, 75% reported having GHG emissions reduction targets, and of those suppliers, 36% reported that their target is science-based. We aim to accelerate our suppliers' GHG emissions reductions through further integration of climate performance into key procurement tools and the development of new mechanisms to drive deep supplier decarbonization.

Advancing carbon-free energy

Our primary approach for reducing our Scope 2 emissions is through the procurement of carbon-free energy.[83] Since 2017, we've matched 100% of the electricity consumption of our global operations with purchases of renewable energy on an annual basis.[84]

However, because of differences in the availability of renewable energy sources like solar and wind across the regions where we operate—and because of the variable supply of these resources—we still need to rely on carbon-emitting energy sources that power local grids (see Figure 15). That's why, in 2020, we set a goal to run on 24/7 carbon-free energy on every grid where we operate by 2030, aiming to procure clean energy to meet our electricity needs, every hour of every day, within every grid where we operate. Achieving this will also increase the impact of our clean energy procurement on the decarbonization of the grids that serve us.[85]

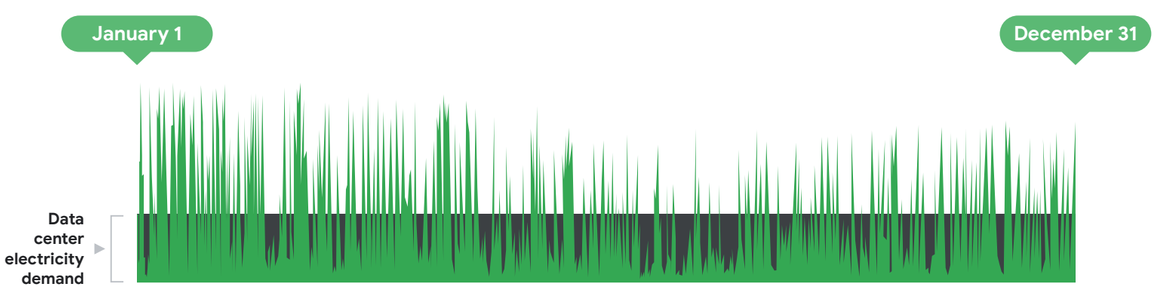
We're working to achieve 24/7 CFE through three main initiatives: **purchasing carbon-free energy**, **accelerating new and improved technologies**, and **transforming the energy system** through partnerships and advocacy.

Purchasing carbon-free energy

Achieving 24/7 CFE is far more complex and technically challenging than annually matching our energy use with renewable energy purchases. No company of our size has achieved 24/7 CFE before, and there's no playbook for making it happen. But we see our efforts as part of a bigger picture, and we've set this ambitious goal to help scale new, global solutions. We're excited to see others—like the U.S. federal government—set similar goals as well.

FIGURE 15

Hourly carbon-free energy performance at an example data center
While Google buys large amounts of wind and solar power (symbolized by green spikes below), these resources are variable, meaning that our data centers still sometimes rely on carbon-based resources.



Renewable energy procurement

We buy electricity directly from new wind and solar farms via long-term PPAs on the grids where we operate, and we also buy renewable power through utilities via renewable energy purchasing models that we helped create.

From 2010 to 2022, we signed more than **80 agreements totaling approximately 10 GW** of clean energy generation capacity—the equivalent of more than 31 million solar panels. Through these agreements, we estimate we'll spend approximately $10 billion to purchase clean energy through 2040.[86]

In 2022, we signed contracts for approximately 2.8 GW of clean energy capacity—more than in any prior year. This included two new projects in the U.K. and Spain, our first in each country, that will help our Cloud regions and offices move closer to 24/7 CFE.

2022 HIGHLIGHT

From 2010 to 2022, we signed more than 80 agreements totaling approximately 10 GW of clean energy generation capacity. Through these agreements, we estimate we'll spend approximately $10 billion to purchase clean energy globally through 2040.[87]



Maevaara wind farm in Sweden (105 MW for Google)

In 2022, we also saw projects we'd contracted over the past few years becoming operational in Belgium, Chile, Denmark, and Finland—providing important clean energy resources to the grids where we operate, and bringing us closer to 24/7 CFE:

- **Belgium:** At our facility in St. Ghislain, Belgium, we installed the first-ever battery-based backup power system for a data center.
- **Chile:** In 2019, Google signed a contract with AES Chile to build 23 new wind turbines in Chile's Biobío region. The project is part of a hybrid wind and solar portfolio that adds 125 MW of clean energy generation capacity. With this wind farm now operational, our data center in Chile operated at 90% CFE in 2022.
- **Denmark:** The Rødby Fjord solar project began adding carbon-free energy to the grid that powers our data center in Fredericia, Denmark. In total, this project adds 54.5 MW of generation capacity to the more than 100 MW of existing operational solar resources in Denmark.
- **Finland:** Google signed a contract with Ilmatar for approximately 60% of the 211 MW generation capacity Piiparinmäki, Finland's largest wind farm, which was completed in November 2021. Google's early agreement to buy the output of this project was critical to the wind farm getting built, and in 2022 it became fully operational and began feeding into the grid that powers our Hamina data center.

In 2017, we became the first major company to match 100% of the annual electricity consumption of our global operations with renewable energy purchases. As of the end of 2022, we've now achieved **six consecutive years** of 100% renewable energy matching on an annual basis (see Figure 16).

From 2011 to 2022, our renewable energy purchasing has resulted in emissions savings of more than 30 million tCO_2e—equivalent to taking more than 6 million cars off the road for a year, or the carbon sequestered by more than 36 million acres of U.S. forests in a year[88] (see Figure 17).



Rødby solar farm in Denmark (55 MW for Google)

FIGURE 16

Renewable energy purchasing compared with total electricity use

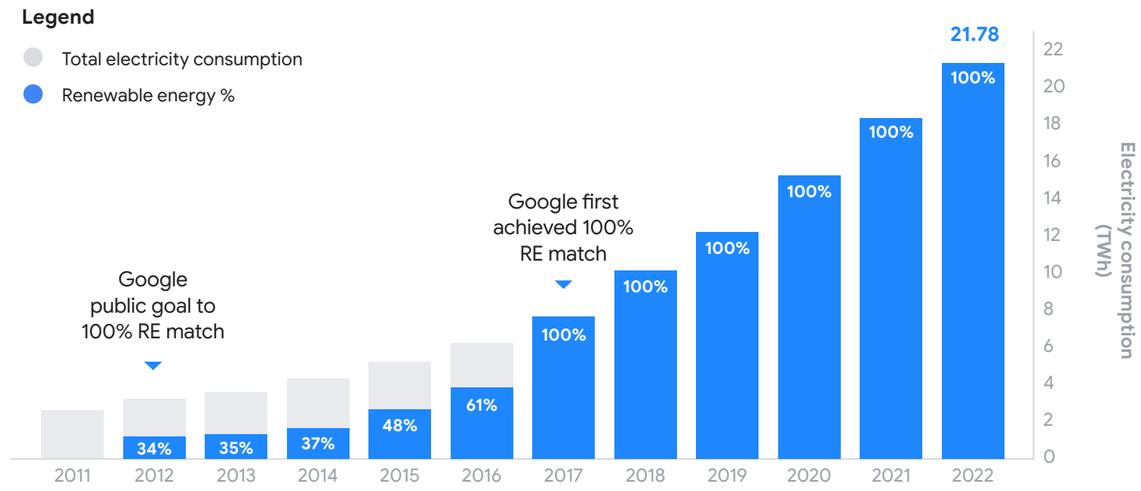
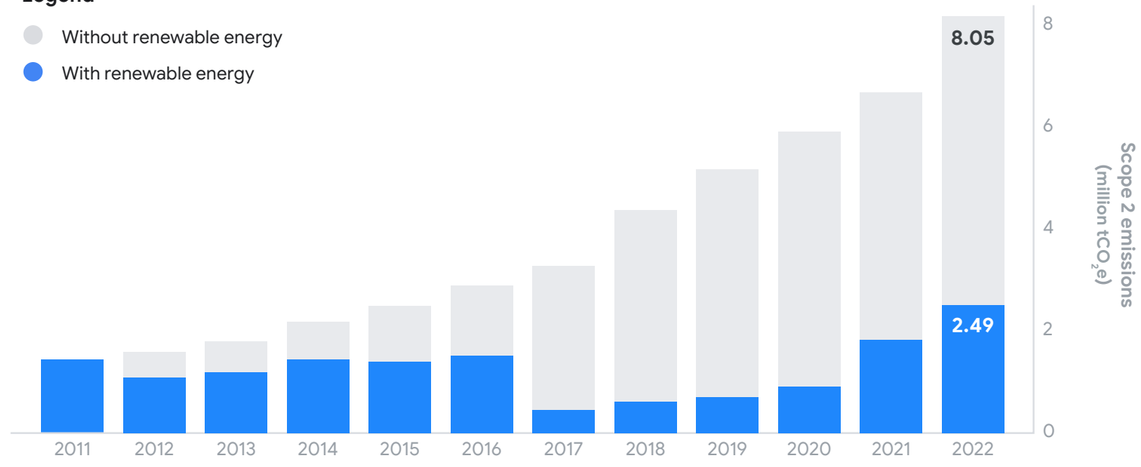


FIGURE 17

Annual impact of renewable energy purchases on Scope 2 emissions



TARGET

Run on 24/7 carbon-free energy on every grid where we operate by 2030

2022 PROGRESS

64% global average carbon-free energy across our data centers and offices

DETAILS

Year set: 2020 | Base year: N/A | Target year: 2030

Scope: Regional % carbon-free energy scores[89], averaged across Google's global portfolio of data centers and offices for 2022. This represents the clean energy procured to meet our electricity needs, every hour of every day, within every grid where we operate. In 2022, we expanded our reporting from only Google-owned and -operated data centers to also incorporate offices and third-party data centers.

24/7 CFE progress

We believe in transparency as a guiding principle, so this year, we're expanding our carbon-free energy reporting. For the first time, we're including CFE scores for both Google-owned and -operated data centers and **third-party facilities** where Google leases space and data processing capacity to support our services. As a result, there are new grid regions[90] included in this year's reporting. We're also including a separate CFE score of the global average for our **offices**.

Data centers

In 2022, we achieved approximately **64%** round-the-clock carbon-free energy across all of our data center sites, inclusive of those operated by third parties. Our 2022 CFE represents a 2% change from 2021 (66% CFE), as a result of updating our metric this year to include third-party operated facilities. Without this change in methodology, CFE for Google-owned and -operated data centers would have remained relatively flat year-over-year.

Our total electricity load across all data center facilities increased by roughly 3 TWh (18%) from 2021 to 2022, while our contracted CFE[91] increased by roughly 2 TWh (17%). Despite this significant expansion of carbon-free energy supply, the percentage of our energy use that came from carbon-free sources remained relatively flat because of the significant growth in our total electricity load.

Beyond the **challenge of growing our CFE procurement** to keep up with our growing power demand, last year we also faced headwinds that made it more challenging to purchase CFE—including continued disruptions to renewable energy supply chains, delays for interconnecting new utility-scale projects, and increased demand for clean energy projects. Adding to these difficulties, load growth continued in regions where CFE contracting is especially difficult given local market dynamics and resource availability constraints (such as the PJM [Pennsylvania-New Jersey-Maryland] grid region in the eastern United States and certain Asia Pacific grid regions).

Despite these significant headwinds, we're proud of the progress we were able to make across many of our data center facilities. **Seven of our 41 grid regions achieved at least 90% CFE**, and 13 achieved at least 85% CFE (see Figure 18). These 13 grid regions span North America, Latin America, and Europe.

While load grew significantly within the central United States, our CFE procurement in this region kept pace—keeping the Midcontinent Independent System Operator and Southwest Power Pool among our highest-scoring grid regions. We increased our contracted CFE in the U.K., Finland, and Germany.

To see more data regarding CFE by region, see the Environmental data table.

Offices

For our offices, which represent a small portion of our global CFE score, we achieved approximately **56%** round-the-clock carbon-free energy.

CFE for commercial real estate faces different challenges, including variability in building characteristics, work patterns, and geographic locations with a broad range of electricity grid carbon intensities. In addition to helping us achieve our own climate goals, pursuing CFE for our offices can help expand clean energy options in the cities we call home.

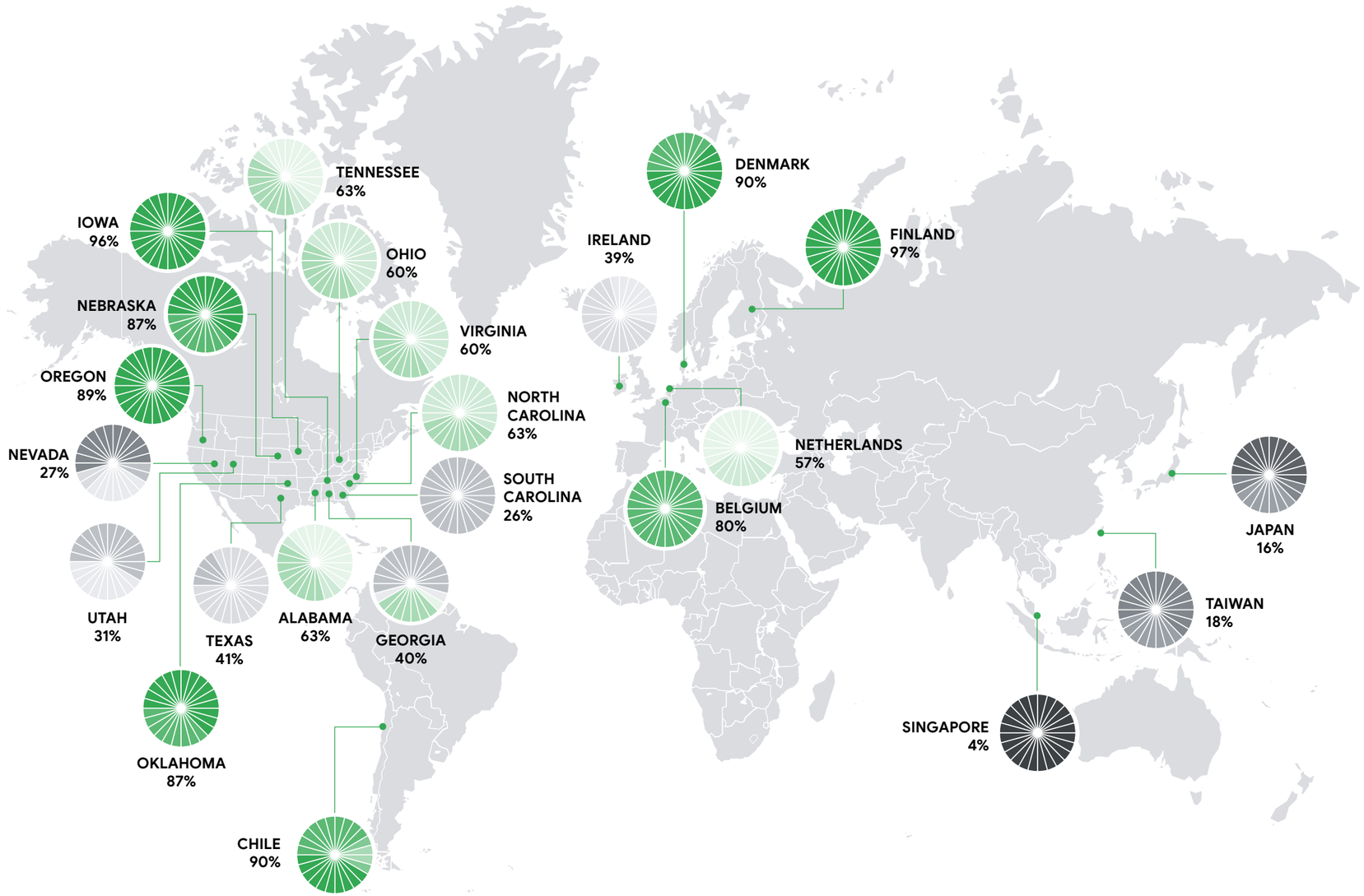


Golden Hills wind farm in California (43 MW for Google)

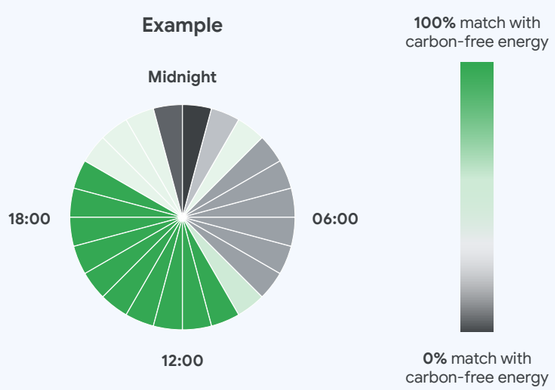
FIGURE 18

Global data center carbon-free energy map

In 2022, Google reached 64% carbon-free energy globally on an hourly basis. This performance varied widely by region, with seven of our regions achieving at least 90% carbon-free energy, and thirteen achieving at least 85%.



How to read clocks



Innovative purchasing models

Our early adoption and long-term support of clean energy projects has coincided with the rapid growth of the industry and remarkable declines in the cost of solar and wind power.[92] Along the way, we've pioneered innovative new contracting models and industry partnerships to help accelerate the adoption of corporate clean energy practices and make clean energy more accessible to a wider range of corporate buyers.

New contracting approaches

In 2022, we published a white paper outlining details of the **CFE Manager model**, a new approach to clean energy procurement that can help a wide range of energy buyers streamline their clean energy purchasing. We share three case studies where Google recently signed CFE Manager agreements—in Virginia, Germany, and California—and discuss some contractual terms and issues that buyers pursuing these agreements should consider.

In addition to the CFE manager model, in 2022 we piloted a new approach to clean energy requests for proposals that can reduce the time to negotiate and execute a clean energy PPA by up to 80%. We're hopeful this new approach will give clean energy buyers and sellers useful new options for negotiating PPAs, and enable all organizations that want to decarbonize their electricity use to join us on the journey to 24/7 CFE.

Helping local communities

Finally, we believe that our clean energy procurement shouldn't only benefit the climate, but also the local communities that host our facilities—in particular, **historically underserved communities** that haven't typically benefited from the clean energy transition. Three years ago, we started a journey to better understand how we can help lower barriers to an equitable clean energy transition, culminating in two additions to our clean energy procurement approach that respond to those challenges.

The first program extends clean energy's financial benefits to local communities. Through a first-of-its-kind partnership with EDPR NA Distributed Generation, we'll create a 500 MW community-based solar portfolio, adding clean energy to the regional U.S. PJM power grid and providing Google with Renewable Energy Credits that contribute to 24/7 CFE. At least 10% of the portfolio's revenues will be redirected annually for up to 15 years as utility bill credits to more than 25,000 households facing a high energy burden.

The second program aims to spur long-term investments to improve housing infrastructure. Through a partnership with Sol Systems, we're providing funding to three rural electric cooperatives and one regional organization with deep roots in North Carolina and South Carolina to undertake critical home pre-weatherization and safety upgrades for low- and moderate-income households. Alongside this investment, the partnership will deliver new solar energy and battery storage resources, moving us closer to 24/7 CFE and bringing clean energy to one of the most challenging grids to decarbonize.

Both of these programs are based on discussions with U.S. government officials, national NGOs, and policy advocates about challenges to an equitable energy transition in their communities. We'll continue to build on these programs to extend the benefits of our investments to underserved communities.

Accelerating new and improved technologies

Wind and solar power have played a critical role in enabling Google's clean energy progress, but meeting 24/7 CFE—and maximizing our contribution to global decarbonization—will require expanding our technology toolkit. In 2022, we continued to work on a wide range of projects to demonstrate, scale, and maximize the climate impact of carbon-free energy technologies, using Google's engineering capabilities and purchasing demand to accelerate the commercialization of new clean energy technologies:

- **Optimizing generation from existing CFE technologies:** To reduce the carbon footprint of our new Bay View and Charleston East campuses, we designed and installed "dragonscale," a new solar roofing technology that helps us power our facilities through on-site generation. Learn more in our Bay View spotlight.
- **Managing energy demand across our global operations:** We developed a "carbon-intelligent computing platform" that optimizes the timing and location of computing tasks based on local CFE availability. This allows us to shift some of our data centers' computing tasks to the times of day when local solar and wind power are most plentiful, and to shift tasks geographically across data centers so that we can do more computing in regions where CFE is abundant.
- **Accelerating next generation energy sources:** In 2021, we signed the first corporate agreement to develop a next-generation geothermal power project, and in 2022, we installed the first-ever battery-based backup power system for a data center at our facility in St. Ghislain, Belgium. This application will enable Google to optimize the carbon footprint of backup power, buying local, low-carbon power for backup when available rather than relying on diesel generators.
- **Improving energy data and clean energy tracking:** Google is driving the development of advanced methods for tracking clean energy, including time-based energy attribute certificates. We're also working to improve the quality, consistency, and granularity of grid electricity data through active collaborations with organizations including EnergyTag, Electricity Maps, FlexiDAO, and more.

> We're deploying machine learning solutions to optimize wind power forecasting and project economics.



For our Bay View campus, a first-of-its-kind building-integrated solar panel called "dragonscale" was developed to create a seamless design. (Photo: Iwan Baan)